This form documents the artifacts associated with the article (i.e., the data and code supporting the computational findings) and describes how to reproduce the findings.

Part 1: Data

- ☐ This paper does not involve analysis of external data (i.e., no data are used or the only data are generated by the authors via simulation in their code).
- ☒ I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Abstract

The proposed model is applied to data from the Parkinson's Progression Markers Initiative (PPMI). The PPMI is a large-scale, longitudinal study designed to collect a comprehensive set of clinical, imaging, biological, genetic, and digital measures across the full spectrum of Parkinson's disease (PD). PPMI uses the Movement Disorder Society-sponsored Unified Parkinson's Disease Rating Scale revision (MDS-UPDRS) to assess PD symptom severity. For model fitting, we primarily use longitudinal MDS-UPDRS scores, baseline demographic covariates, and putamen volume. For model validation, we incorporate auxiliary variables from PPMI, including medication usage, SCOPA-AUT scores, REM Sleep Behavior Disorder scores, and clinician diagnoses.

Availability

- ☒ Data **are** publicly available.
- ☐ Data **cannot be made** publicly available.

If the data are publicly available, see the *Publicly available data* section. Otherwise, see the *Non-publicly available data* section, below.

Publicly available data

- ☐ Data are available online at:
- ☐ Data are available as part of the paper's supplementary material.
- ☒ Data are publicly available by request, following the process described here: https://www.ppmi-info.org/access-data-specimens/download-data
- ☐ Data are or will be made available through some other mechanism, described here:

Non-publicly available data

Description

File format(s)

- ☒ CSV or other plain text.
- ☐ Software-specific binary format (.Rda, Python pickle, etc.): pkcle
- ☐ Standardized binary format (e.g., netCDF, HDF5, etc.):
- ☐ Other (please specify):

Data dictionary

- ☐ Provided by authors in the following file(s):
- ☐ Data file(s) is(are) self-describing (e.g., netCDF files)
- ☒ Available at the following URL: https://www.ppmi-info.org/access-data-specimens/download-data (under the "Study Docs" category).

Additional Information (optional)

Data used in this article was obtained on [2023-01-03] from the Parkinson's Progression Markers Initiative database (https://www.ppmi-info.org/access-data-specimens/download-data), RRID:SCR_006431. Because the PPMI repository undergoes continuous updates and direct data sharing is prohibited under their Data Use Agreement and Publication Policy, we have generated and provided a synthetic dataset. This ensures the long-term computational reproducibility of our code.

Part 2: Code

Abstract

We provide the code to implement our proposed Longitudinal Multi-Domain Latent Structure Model. Please refer to README.md for details

Description

Code format(s)

- ☒ Script files
  - ☒ R
  - ☐ Python
  - ☐ Matlab
  - ☒ Other: cpp
- ☐ Package
  - ☐ R
  - ☐ Python
  - ☐ MATLAB toolbox
  - ☐ Other:
- ☐ Reproducible report
  - ☐ R Markdown
  - ☐ Jupyter notebook
  - ☐ Other:
- ☒ Shell script
- ☐ Other (please specify):

Supporting software requirements

Version of primary software used R version 4.3.3; C++ Compiler: g++ version 13.3.0

Libraries and dependencies used by the code Rcpp (v1.1.1); RcppArmadillo (v15.2.3.1); RcppParallel (v5.1.11.1); MCMCpack (v1.7.1); MASS (v7.3.60.0.1); dqrng (v0.4.1); doParallel (v1.0.17); doRNG (v1.8.6.2); mclust (v6.1.2); clue (v0.3.66); coda (v0.19.4.1); ggplot2 (v4.0.2); patchwork (v1.3.2); gridExtra (v2.3); dplyr (v1.1.4); tidyr (v1.3.2); reshape2(v1.4.5); vcd (v1.4.13)

Supporting system/hardware requirements (optional)

Linux (tested on Ubuntu 24.04 LTS). Note: Running on Windows or macOS may yield different numerical results due to platform differences in floating-point arithmetic, thread scheduling, and random number generation.

Parallelization used

- ☐ No parallel code used
- ☒ Multi-core parallelization on a single machine/node

- Number of cores used: 500

☐ Multi-machine/multi-node parallelization
- Number of nodes and cores used:

License

☒ MIT License (default)
☐ BSD
☐ GPL v3.0
☐ Creative Commons
☐ Other: (please specify)

Additional information (optional)

Part 3: Reproducibility workflow

Scope

The provided workflow reproduces:

☐ Any numbers provided in text in the paper
☒ The computational method(s) presented in the paper (i.e., code is provided that implements the method(s))
☒ All tables and figures in the paper
☐ Selected tables and figures in the paper, as explained and justified below:

Workflow

Location

The workflow is available:

☐ As part of the paper's supplementary material.
☐ In this Git repository:
☒ Other (please specify): The code and workflow are available in the uploaded files.

Format(s)

☐ Single master code file
☒ Wrapper (shell) script(s)
☐ Self-contained R Markdown file, Jupyter notebook, or other literate programming approach
☒ Text file (e.g., a readme-style file) that documents workflow
☐ Makefile
☐ Other (more detail in *Instructions* below)

Instructions

See README.md for detailed steps.

Expected run-time

Approximate time needed to reproduce the analyses on a standard desktop machine:

☐ < 1 minute
☐ 1-10 minutes
☐ 10-60 minutes
☐ 1-8 hours
☒ > 8 hours

☐ Not feasible to run on a desktop machine, as described here:

Additional information (optional)

Notes (optional)